

Library Usage Data Warehouse Dashboard Guide

1. Introduction

The **Operational Dashboard** provides a **real-time overview** of the daily operations of the University Library. It integrates data from three systems:

1. **Book Lending System** – Physical book checkouts and returns
2. **Digital Resources System** – E-book, journal, and article downloads
3. **Study Room Booking System** – Room reservations

This dashboard is designed for **library staff and department administrators** to monitor library usage, identify trends, and make data-driven operational decisions.

2. Purpose of the Dashboard

- Track **daily usage metrics** across all library services
- Identify **peak usage times** for study rooms and digital resources
- Monitor **resource efficiency** (e.g., loan durations, digital resource engagement)
- Provide actionable insights for **staff allocation** and **resource planning**
- Ensure **data-driven decision-making** with interactive visualizations

3. Dashboard Layout & Components

The operational dashboard is divided into **three main sections**:

A. Summary Metrics (Top KPIs)

- **Total Books Checked Out Today:** Number of books borrowed
- **Total Digital Downloads:** Number of e-books, journals, and articles accessed
- **Room Bookings Today:** Number of study room reservations
- **Average Loan Duration:** Average number of days books are borrowed

B. Daily Trends & Heatmaps

- **Book Checkouts by Department:** Bar chart showing checkouts per department
- **Digital Downloads Over Time:** Line chart for hourly or daily downloads
- **Room Booking Occupancy:** Heatmap showing room usage by hour and room number

C. Operational Alerts & Highlights

- **Late Returns:** List of overdue books with student IDs (masked for privacy)
- **Unassigned Bookings:** Room bookings missing student ID
- **High-Demand Resources:** Books or digital resources with high usage today

4. Interactive Features

- **Date Selector:** Filter data for a specific date or date range
- **Department Filter:** View metrics for a particular department
- **Resource Type Filter:** Select Books, E-books, Journals, or Articles
- **Room Selector:** Drill down into specific study rooms
- **Drill-Down & Roll-Up:** View data from daily → weekly → monthly levels
- **Pivot Views:** Compare usage across departments and resource types

5. How to Use

1. **Open the Dashboard** using Excel, Power BI, Tableau, or the provided file format
2. **Select Date/Time Range:** Use the date filter to focus on specific days
3. **Apply Department or Resource Filters:** Refine results based on user requirements
4. **View KPIs and Trends:** Monitor the top metrics and visualizations
5. **Analyze Alerts:** Check for overdue returns or unassigned bookings
6. **Export Data:** If needed, export charts and tables for reporting

6. Security & Access

- **Role-Based Access Control (RBAC)** ensures only authorized users can view sensitive data
 - **Admin:** Full access
 - **Department Head:** Access only to department-specific data
 - **Staff/Analyst:** Read-only access to operational metrics
- **Student IDs** are masked to protect privacy

- **Data Refresh:** Dashboard updates **daily** from the data warehouse

7. Maintenance & Troubleshooting

- **Data Refresh Errors:** Check ETL logs if data seems outdated
- **Filter Not Working:** Ensure correct filter selection and refresh the dashboard
- **Missing Metrics:** Verify that source data files are correctly loaded in the staging area

8. Additional Notes

- The operational dashboard **complements** the Executive and Department Dashboards by focusing on **day-to-day activities**
- Regularly monitor **peak usage times** to optimize library operations
- Use insights for **resource allocation, room scheduling, and digital resource planning**